Software Modeling and Design Project

Topic: Food Ordering and Delivering System
Group: A
Members: Erjus Allka,Arjola Sula,Xhesika Toska,Sofia Cala,Klara Hysenaj,Aurelio Hysenaj,Imerr Meraj

User Levels

Primary Users

1. Customer

The customer is the main user who orders food through the application. Customers use the system to browse restaurants, select food items, place orders, and track their deliveries.

Responsibilities:

• Register and log into the system
• Browse restaurants and food menus
• Search for restaurants or food items
• Add food items to the cart
• Place food orders
• Make payments for orders
• Track order and delivery status
• View order history
• Rate and review restaurants and food

2. Restaurant Staff

Restaurant staff manage food orders received from customers and prepare the food for delivery.

Responsibilities:

• Log into the system
• Manage restaurant profile information
• Add, update, or remove menu items
• Receive customer orders
• Accept or reject orders
• Prepare food orders
• Update order status (accepted, preparing, ready)
• Manage food availability and pricing

3. Delivery Driver

The delivery driver is responsible for delivering food orders from restaurants to customers.

Responsibilities:

• Log into the system
• View assigned delivery requests
• Accept delivery tasks
• Pick up food orders from restaurants
• Deliver orders to customers
• View delivery addresses and order details
• Update delivery status (picked up, on the way, delivered)
• Confirm order delivery

Secondary Users

4. Administrator (Admin)

The administrator manages the platform and ensures that the system operates correctly.

Responsibilities:

• Manage customer accounts
• Manage restaurant accounts
• Manage delivery driver accounts
• Monitor system activity
• Manage orders and platform data
• Handle user complaints and reports
• Generate system reports
• Ensure system security and reliability

5. Finance Manager

The finance manager manages all financial operations related to the platform.

Responsibilities:

• Monitor customer payments
• Manage financial transactions
• Track platform revenue
• Generate financial reports
• Verify payment records
• Handle payment-related issues

6. Customer Support

Customer support assists users when they encounter problems while using the system.

Responsibilities:

• Respond to customer inquiries
• Assist users with order problems
• Resolve payment issues
• Handle complaints and feedback
• Communicate with restaurants and delivery drivers
• Provide help for using the application

7. System Administrator / IT Support

The system administrator manages the technical infrastructure and ensures the system runs smoothly.

Responsibilities:

• Maintain system servers and databases
• Monitor system performance
• Ensure system security
• Perform system updates and maintenance
• Fix technical issues and system errors
• Manage backups and data recovery

External Users

8. Payment System

The payment system is an external service that processes online payments made by customers. It ensures secure financial transactions between the customer and the platform.

Responsibilities:

• Process online payments for orders
• Verify payment information securely
• Confirm successful or failed transactions
• Handle payment authorization and validation
• Communicate payment results to the platform

9. Notification System

The notification system is responsible for sending system messages and updates to users. It helps inform customers, restaurants, and delivery drivers about important events in the system.

Responsibilities:

• Send order confirmation notifications
• Send order status updates
• Send delivery status notifications
• Send system alerts and reminders
• Notify users about successful or failed payments

2. Functional Requirements

Customer

1. Customer registers an account
2. Customer logs into the system
3. Customer updates personal profile information
4. Customer browses restaurants
5. Customer searches restaurants by name or category
6. Customer views restaurant menus
7. Customer views food item details
8. Customer adds food items to the cart
9. Customer removes food items from the cart
10. Customer updates item quantity in the cart
11. Customer places a food order
12. Customer selects delivery address
13. Customer selects payment method
14. Customer makes online payment
15. Customer tracks order status
16. Customer receives order notifications
17. Customer views order history
18. Customer cancels an order
19. Customer rates restaurants
20. Customer writes reviews for restaurants

Restaurant Staff

21. Restaurant staff logs into the system
22. Restaurant staff manages restaurant profile information
23. Restaurant staff adds menu items
24. Restaurant staff updates menu items
25. Restaurant staff removes menu items

26. Restaurant staff manages food availability
27. Restaurant staff receives customer orders
28. Restaurant staff accepts customer orders
29. Restaurant staff rejects customer orders
30. Restaurant staff updates order preparation status
31. Restaurant staff marks orders as ready for delivery

Delivery Driver

32. Delivery driver logs into the system
33. Delivery driver views assigned delivery requests
34. Delivery driver accepts delivery requests
35. Delivery driver views delivery details
36. Delivery driver navigates to restaurant location
37. Delivery driver picks up food orders from restaurants
38. Delivery driver navigates to customer location
39. Delivery driver delivers food orders to customers
40. Delivery driver updates delivery status
41. Delivery driver confirms order delivery

Administrator

42. Administrator manages customer accounts
43. Administrator manages restaurant accounts
44. Administrator manages delivery driver accounts
45. Administrator monitors system activity
46. Administrator manages orders in the system
47. Administrator handles complaints and reports
48. Administrator generates system reports

Finance Manager

49. Finance manager monitors payment transactions
50. Finance manager generates financial reports

3. Non-Functional Requirements

Performance

1. Performance Requirement
The system shall process user requests and display results within 2 seconds under normal operating conditions.

2. Response Time Requirement
Restaurant menus and food items shall be displayed within 3 seconds after a user request.

3. Concurrent Users Requirement
The system shall support at least 500 concurrent users placing orders simultaneously.

4. System Capacity Requirement
The system shall be capable of handling thousands of orders per day without performance degradation.

5. Notification Performance Requirement
The system shall send order confirmation and delivery notifications within a few seconds after order status changes.

6. Order Tracking Requirement
The system shall update order and delivery status in real time or near real time.

Security Requirements

7. Security Requirement
The system shall protect user accounts and payment information through secure authentication mechanisms and encrypted data storage.

8. Authentication Requirement
The system shall require users to authenticate (register and log in) before accessing personalized services.

9. Authorization Requirement
The system shall restrict access to system functionalities based on user roles such as Customer, Restaurant Staff, Delivery Driver, and Administrator.

10. Privacy Requirement
The system shall ensure that user personal data is confidential and protected from unauthorized access.

11. Payment Security Requirement
The system shall ensure secure communication with external payment systems during financial transactions.

Reliability and Data Management

12. Reliability Requirement
The system shall ensure that all orders are processed correctly without duplication or loss.

13. Data Integrity Requirement
The system shall maintain accurate and consistent data for users, orders, and payment transactions.

14. Backup Requirement
The system shall perform automatic database backups on a daily basis to prevent data loss.

15. Recovery Requirement
The system shall be able to restore system data from backups in case of system failure.

16. Data Storage Requirement
The system shall securely store user profiles, order history, and transaction records.

Usability and Accessibility

17. Usability Requirement
The system interface shall be simple, intuitive, and easy to navigate for all types of users.

18. Accessibility Requirement
The system shall be accessible through web browsers and mobile devices with an internet connection.

19. Error Handling Requirement
The system shall display clear and informative error messages when problems occur.

Compatibility and Scalability

20. Compatibility Requirement
The system shall function correctly on major web browsers such as Chrome, Firefox, and Edge.

21. Scalability Requirement
The system shall support growth in users, restaurants, and delivery drivers without performance degradation.

Maintainability and Monitoring

22. Maintainability Requirement
The system shall be developed using a modular architecture to simplify system maintenance and updates.

23. Logging Requirement
The system shall maintain activity logs of user actions, orders, and system operations.

24. System Monitoring Requirement
The administrator shall be able to monitor system activity and platform performance.

Availability

25. Availability Requirement
The system shall be available 24 hours a day, 7 days a week, with a minimum uptime of 99%.